Investigating the Impact of AI Technologies on Fraudulent Amazon Reviews Using Similarity Analysis

Big Data and Cloud Computing Final Project

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Executive Summary

- **Objective**: To analyze review data from Amazon, and determine if Generative AI has led to increased review fraud, and to evaluate the effectiveness of LSH (Locality Sensitive Hashing) detection.
- **Category Focus**: Arts, Crafts & Sewing.
- **Key Findings**:
  - **Volatility**: Significant review volume spikes observed in late 2019 and 2021.
  - **Uniqueness**: Titles are highly repetitive (46.7%), while texts is relatively unique (9.2%).
  - **AI Impact**: Duplication rates **dropped significantly** in the post-2022 era.
- **Implication**: Generative AI has greatly diversified review text, so traditional similarity detection like LSH seems out of date. New strategies, such as behavioral analysis, should be considered.

Data Pipeline & Methodology Overview

Dataset

- **Source**: Amazon Reviews Archive (Google Cloud Storage)
- **Scale**: Big Data (Parquet format /50+ GB)

Tech Stack

- **Environment**: Google Colab
- **Processing Methods**: PySpark and Pandas
- **Algorithm**: LSH (Jaccard Similarity Threshold $= 0.7$)

Data Cleaning Funnel:

1. Transported batch to Colab SSD to bypass memory limits.
2. Standardized timestamps and filtered invalid cols.
3. Inner Joined review and meta data via `parent_asin`.
4. Established a processed clean dataset with than 600M reviews.

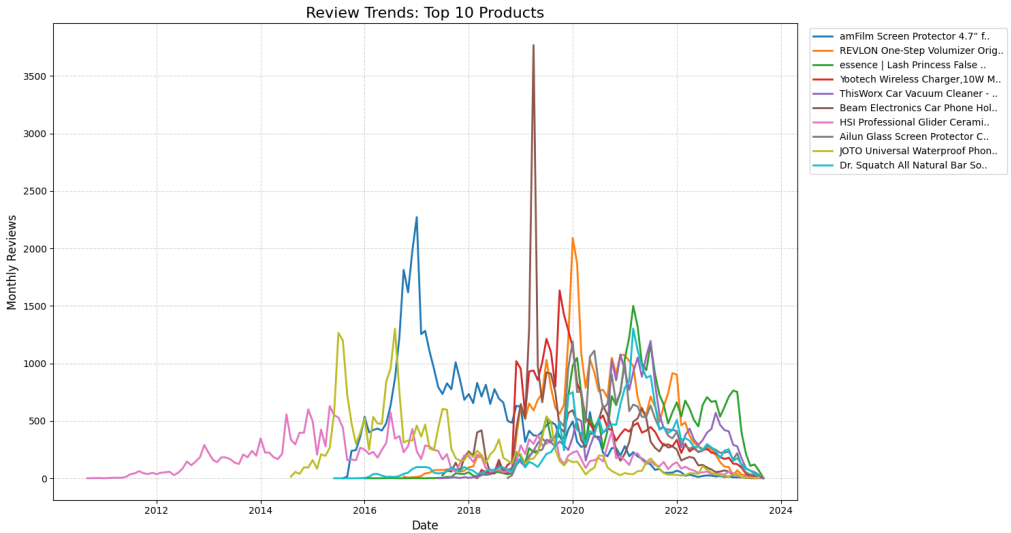
Timeline Analysis: Volatility & Spikes



Observation:

- 2012-2019: Stable growth.
- Late 2019: Massive peak (Likely caused by Pandemic)
- 2021+: Extreme volatility with rapid spikes and drops.

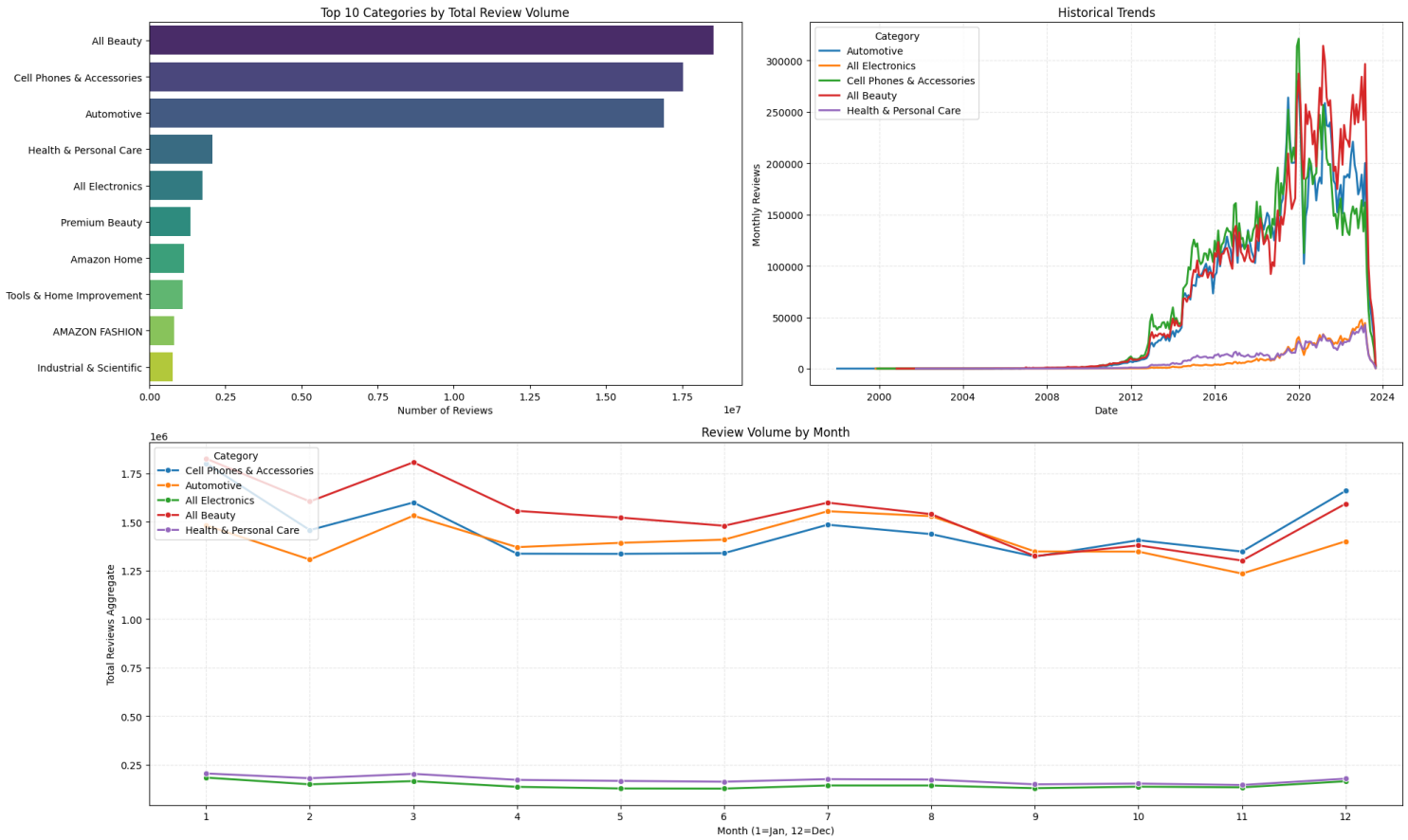
Top 10 Products



Insight:

- **Review Pattern**: Most top products follow the same pattern: they gained an extremely large amount of reviews (went viral) in a short period.
- **Spike Similarity**: Some top products contain a similar spike pattern.

Product Categories



Insight: Review volume is heavily concentrated in top 3 categories (Beauty, Cellphone, and Automotive), with a very weak seasonality, and the spike usually occurs on January and March.

Rating vs. Review Volume Analysis

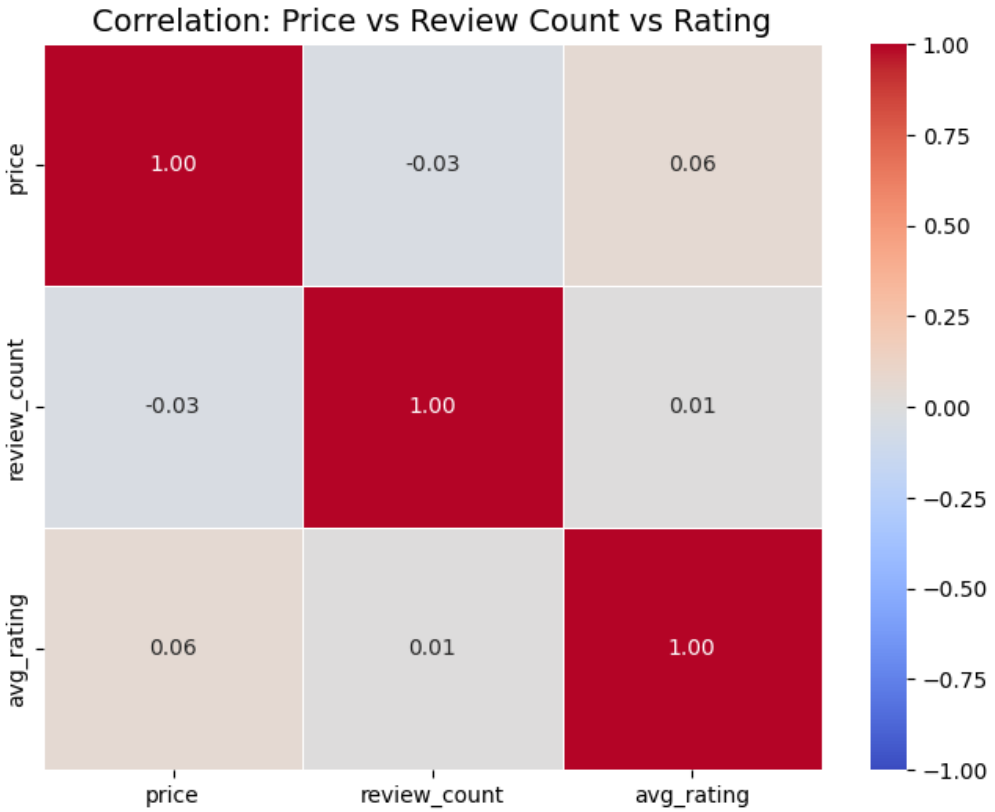
Rating Bucket	Avg Reviews	of Products
4.9 - 5.0 (Ex. High)	12.18	62,850
4.5 - 4.8	45.75	277,360
4.0 - 4.4	**67.99**	378,604
$<$ 4.0	45.76	408,601

Table 1: Average Reviews per Product by Rating Bucket

Insight:

- Products with "Extremely High" scores (4.9-5.0) have the **lowest** review volume (12.18).
- The most reviewed products fall in the 4.0 - 4.4 range.

Price Correlations



Insight:

- **Price vs. Rating**: Weak positive correlation (**0.06**). A higher price does not guarantee (that much) satisfaction.
- **Price vs. Volume**: Very weak negative correlation (**-0.03**). Price is not a barrier for comments.

Common Words (5-Star vs 1-Star)

5-Star Reviews (Positive)

1. **great** (8.7M)
2. **love** (6.7M)
3. **good** (5.0M)
4. **like** (5.0M)
5. **easy** (4.8M)

General satisfaction.

1-Star Reviews (Negative)

1. **one** (1.1M)
2. **product** (1.1M)
3. **like** (1.0M)
4. **phone** (1.0M)
5. **get** (0.8M)

Specific failures ("one", "product").

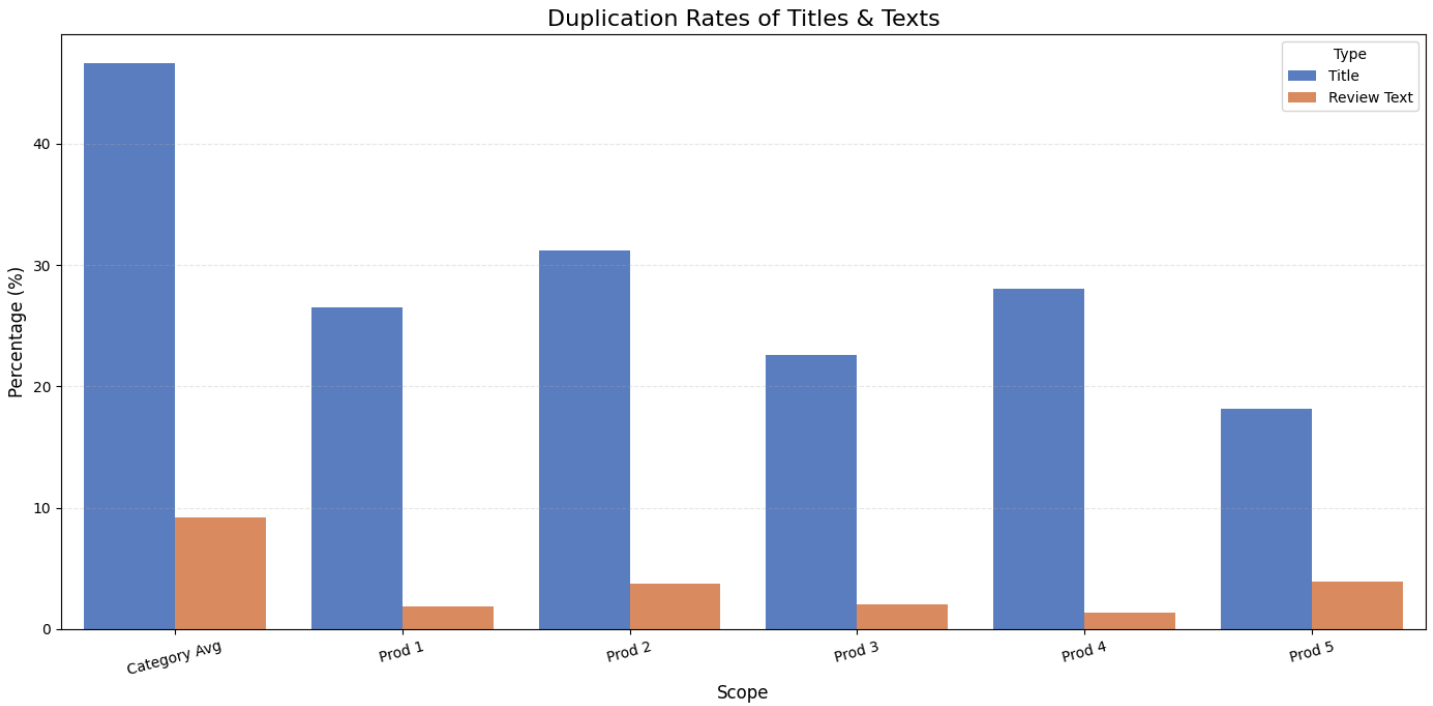
Top Reviewers

- **Top Reviewer** posted **3,400** reviews across **24** categories.
- **Behavior Pattern**:
  - The top 10 reviewers all posted in **13-24 categories**.
  - High volume and high category diversity indicate professional (paid) reviewer or bot account.

Rank	Total Reviews	Unique Categories
#1	3,400	24
#2	2,353	14
#3	2,293	14
#4	1,987	16
#5	1,895	13

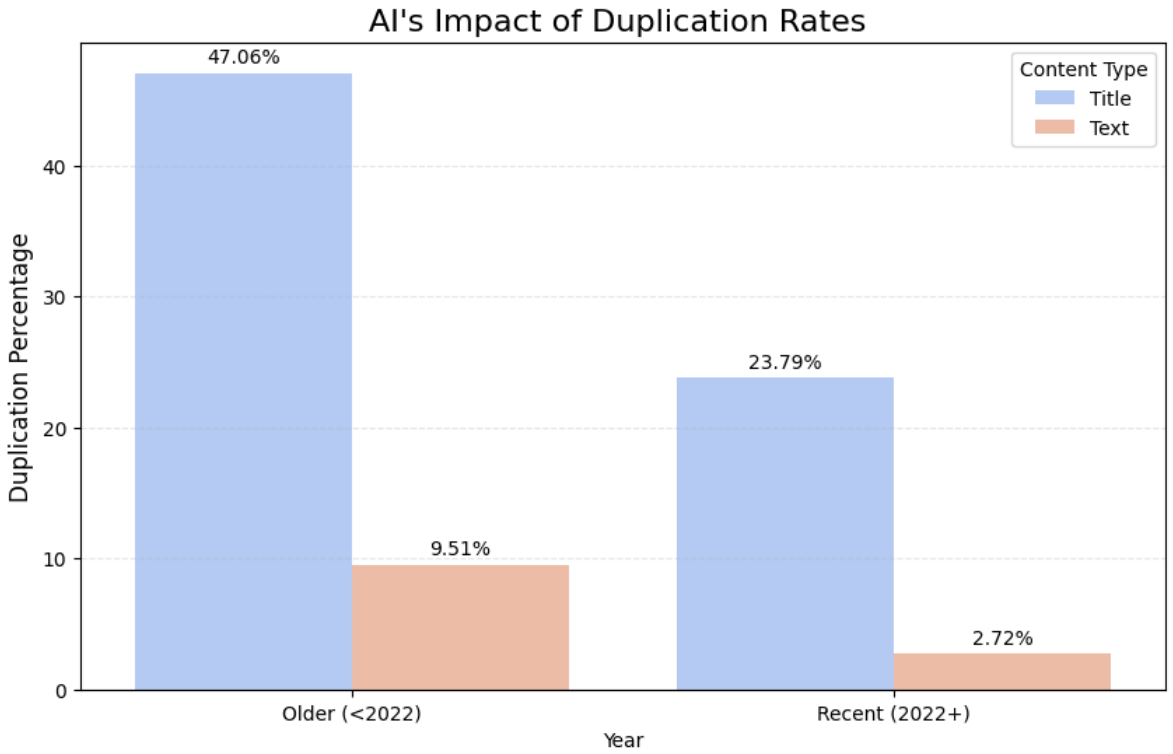
Table 2: Top 5 Active Reviewers

Uniqueness of Arts, Crafts & Sewing: Titles vs. Texts



- **Titles** have high duplicate rate (46.67%), likely due to generic phrases ("Good", "Five Stars").
- **Texts** have a Relatively low rate (9.16%).
- **Top products** have significantly lower rates compared to the category average.

The AI Impact



- Duplication rates dropped significantly in AI era.
- **Conclusion**: Old "Copy-Paste" bots are likely to be replaced by AI tools, which bypasses traditional similarity detection.

Conclusion & Recommendations

The Drop of Detection Effectiveness

The drop in duplication rates indicates more False Negative. It does not mean that spam reviews are gone, and it suggests that spam may have evolved beyond text similarity.

Actionable Recommendations:

1. **Shift Away from Text Similarity**: Traditional methods are becoming less effective against AI-generated reviews.
2. **Focus on Behavioral Metadata**: Try to find ways to detect fraud based on account activity patterns (e.g., review frequency, cross-category posting diversity, purchase history).